

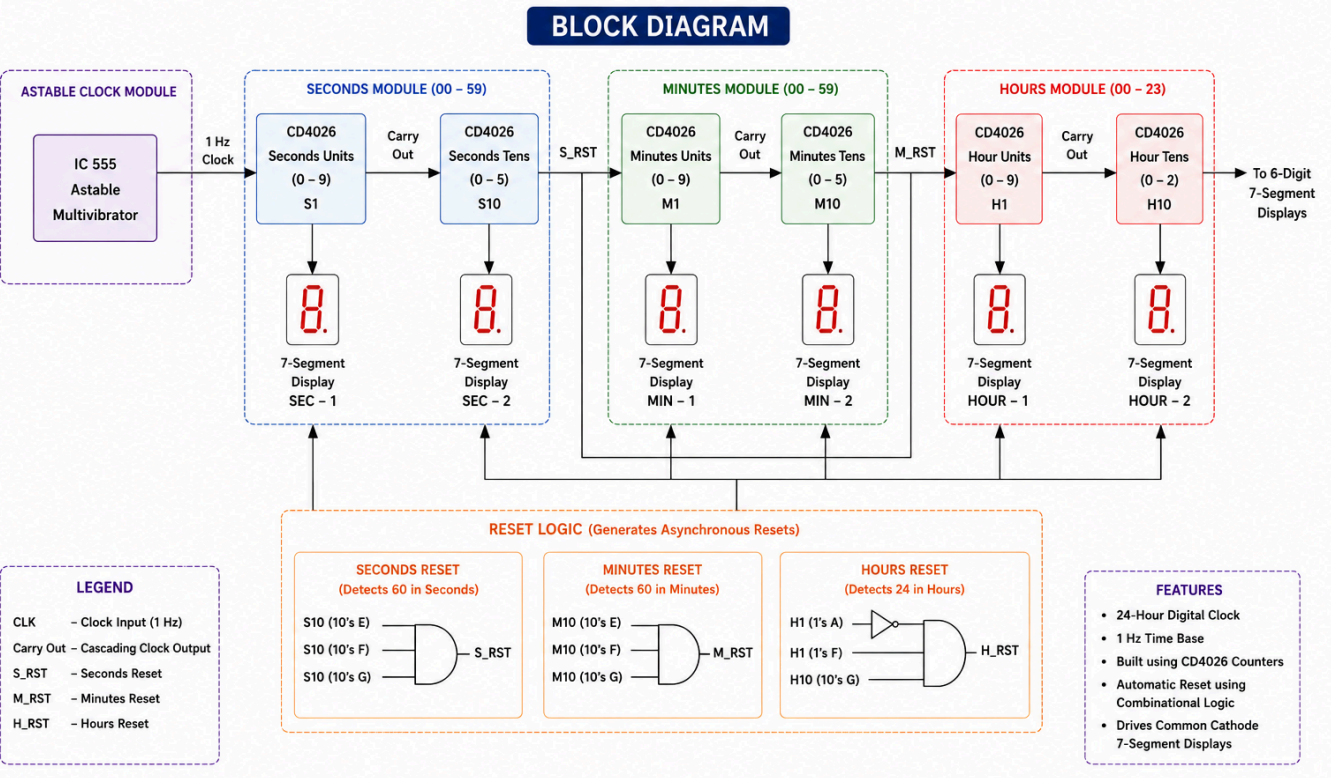
# 24-HOUR DIGITAL CLOCK

## CHRONOBITS'26

This project implements a 24-Hour Digital Clock using the NE555 Timer and CD4026 Counter ICs without a microcontroller. The NE555 generates a 1 Hz clock pulse, while cascaded CD4026 ICs count the pulses and drive six seven-segment displays to show time in HH:MM:SS format. External reset logic is used to achieve MOD-60 and MOD-24 counting for accurate timekeeping.

- Objectives**
- Design a 24-hour digital clock using discrete ICs.
  - Generate a 1 Hz clock pulse using the NE555 Timer.
  - Display time in HH:MM:SS format using CD4026 ICs.
  - Implement MOD-60 and MOD-24 counting using reset logic.

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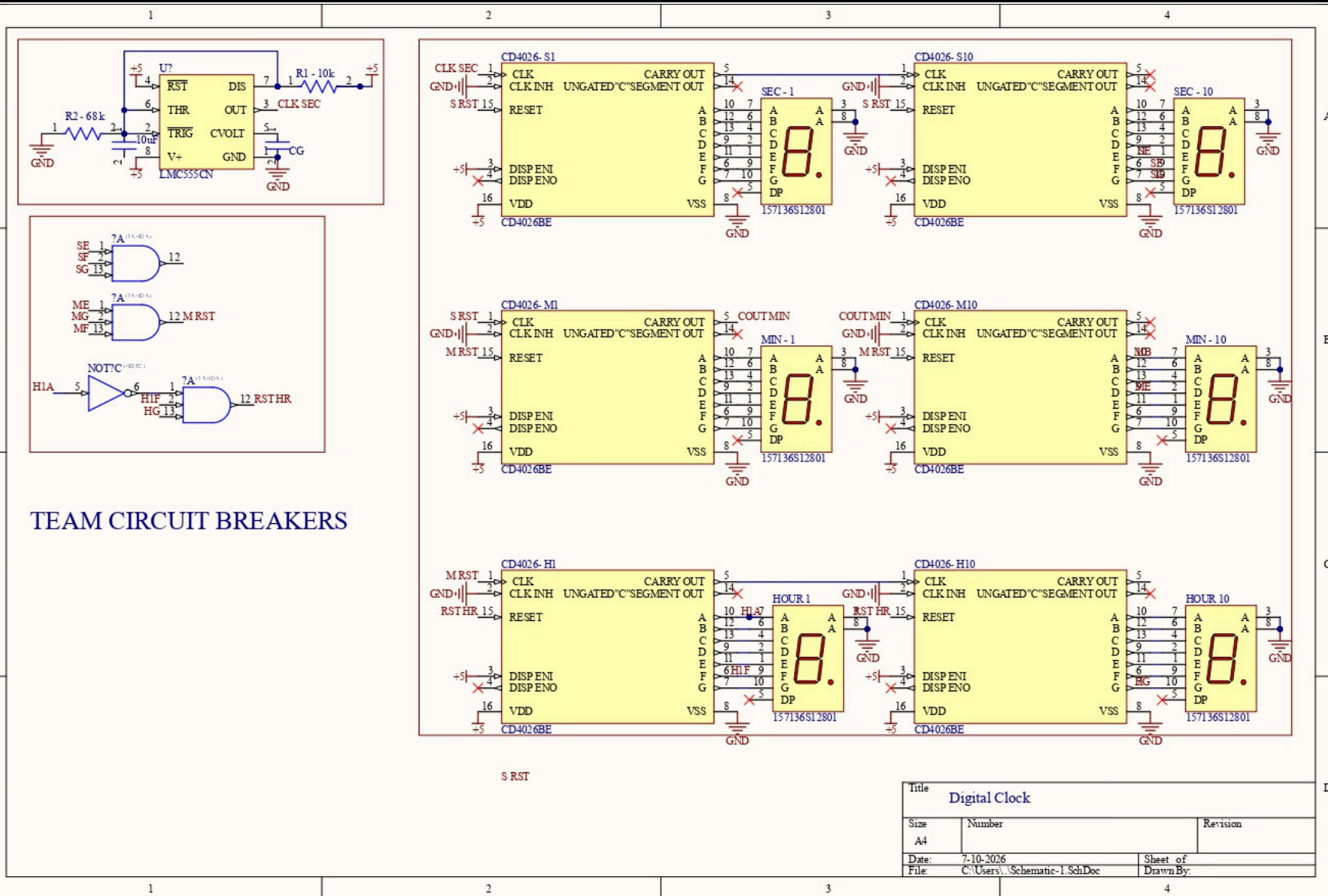
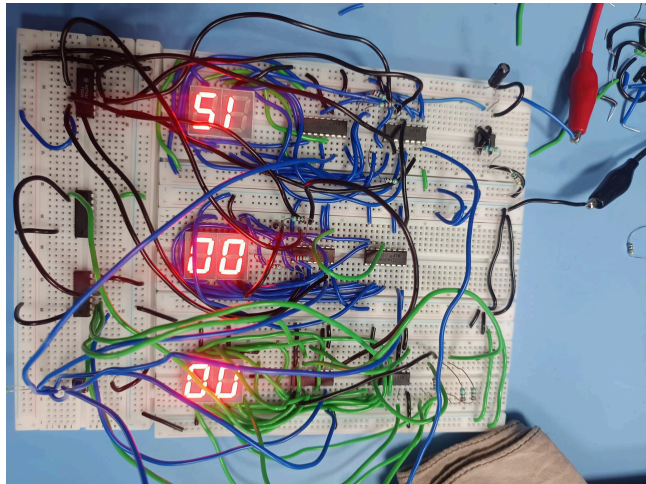


### Working Principle

- The NE555 Timer is configured in astable mode to generate a continuous 1 Hz clock pulse.
- Each pulse is applied to the first CD4026 counter, which increments the seconds display.
- The Carry Out output cascades the count to the next CD4026, forming the seconds, minutes, and hours counters.
- External reset logic detects the count of 60 for seconds and minutes, resetting them to 00 (MOD-60).
- Additional logic detects 24 hours and resets the hour counters to 00, achieving 24-hour operation.
- The time is displayed in HH:MM:SS format using six common-cathode seven-segment displays.
- This version is concise, technically accurate, and fits well on an A3 engineering poster.

### Results

- Successfully displays time in HH:MM:SS
- Correct MOD-60 counting
- Correct MOD-24 rollover
- Stable 1 Hz pulse generation



| Component         | Qty |
|-------------------|-----|
| NE555             | 1   |
| CD4026            | 6   |
| 7-Segment Display | 6   |
| 74HC04            | 1   |
| 74HC11            | 1   |

